

COS30045

DATA VISUALISATION

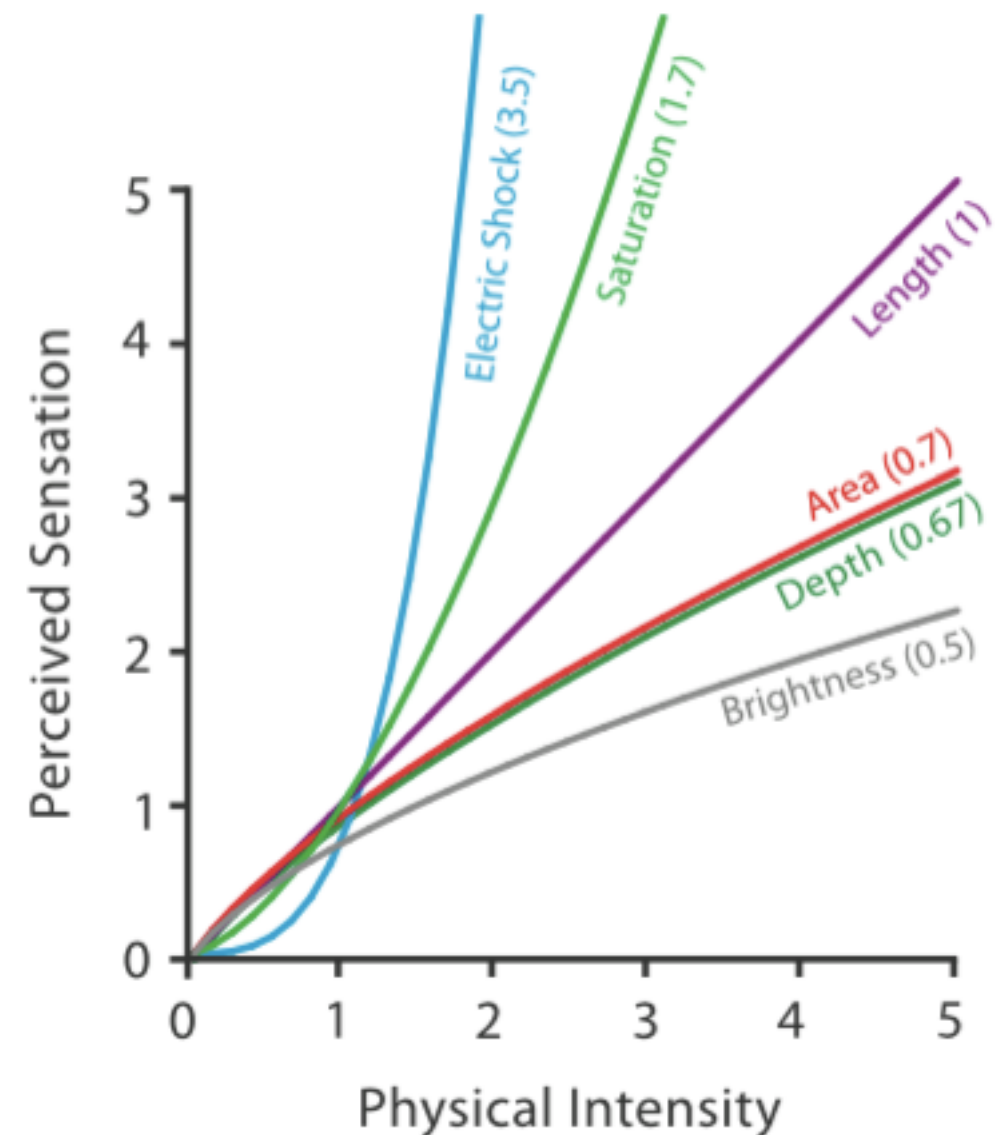
TOPIC 02-4: DESIGN GUIDELINES 3D

GUIDELINES – MUNZNER

Avoid Unjustified 3D

- ▶ Depth judgment is poor
- ▶ Occlusion
- ▶ Perspective Distortion
- ▶ Colour: Lighting / Shadows / Shading
- ▶ Tilted Text illegible

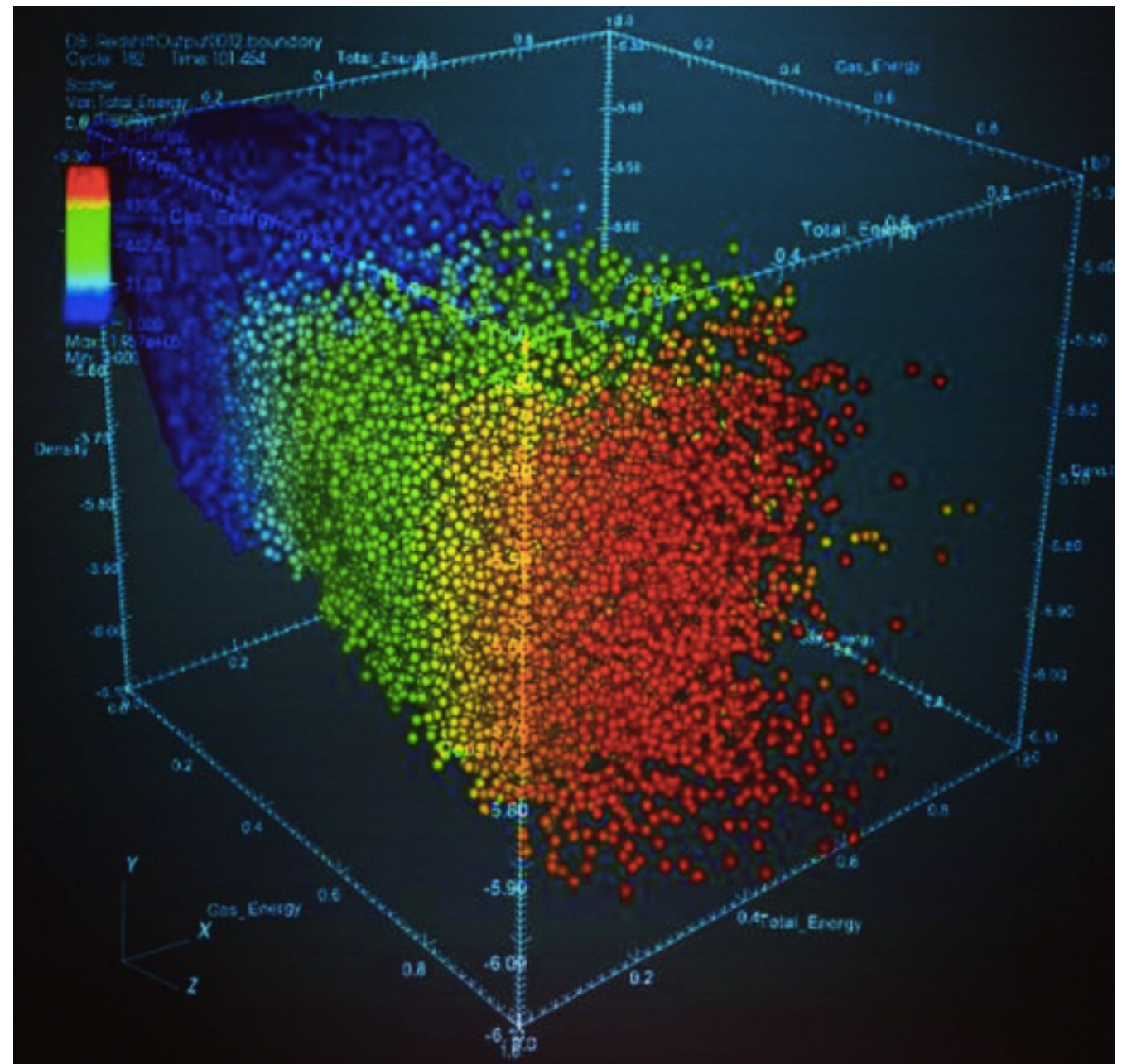
Steven's Psychophysical Power Law: $S = I^N$



GUIDELINES – MUNZNER

Avoid Unjustified 3D

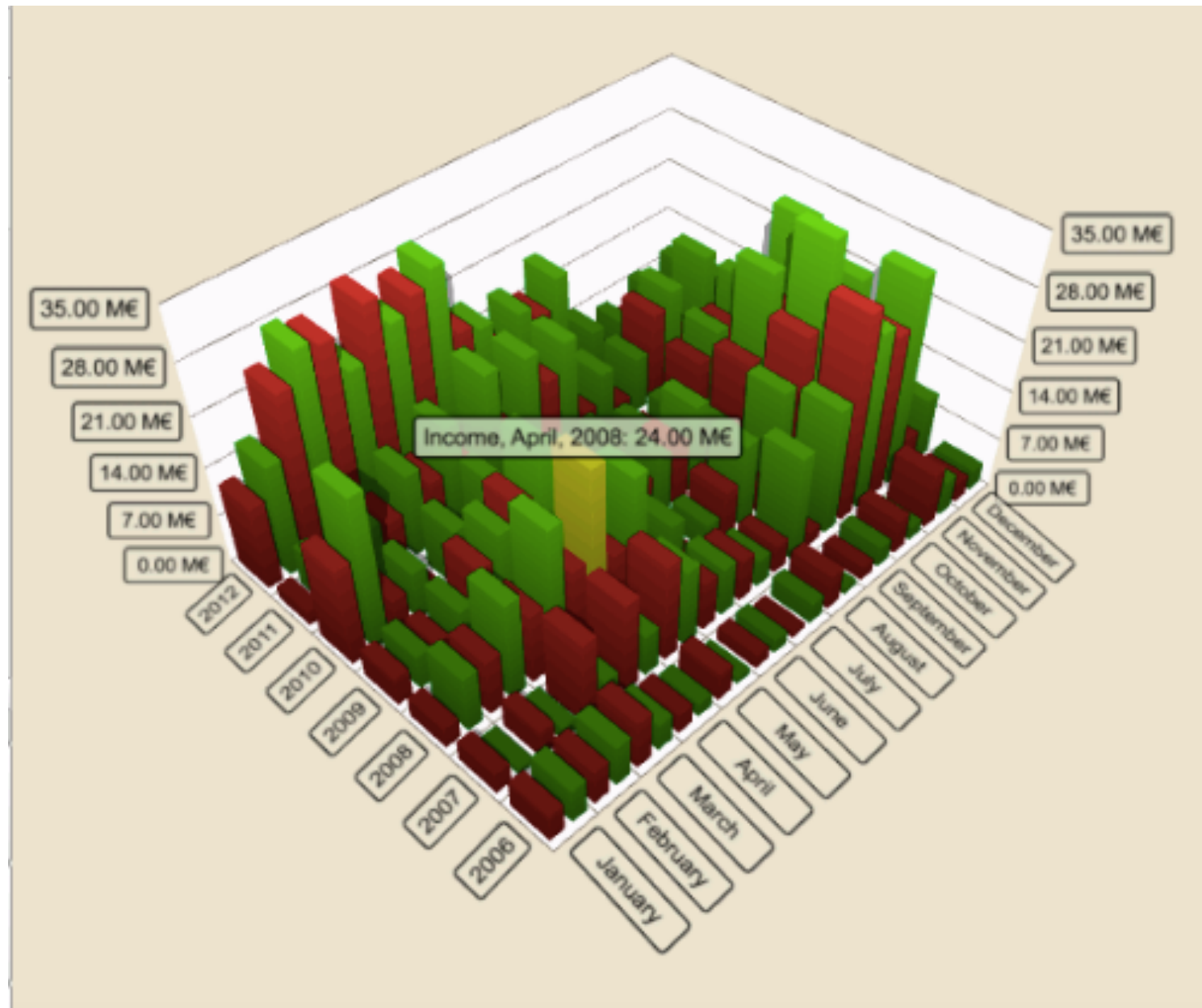
- ▶ Depth judgment is poor
- ▶ Occlusion
- ▶ Perspective Distortion
- ▶ Colour: Lighting / Shadows / Shading
- ▶ Tilted Text illegible



GUIDELINES – MUNZNER

Avoid Unjustified 3D

- ▶ Depth judgment is poor
- ▶ Occlusion
- ▶ Perspective distortion
- ▶ Colour: Lighting / Shadows / Shading
- ▶ Tilted Text illegible



GUIDELINES

Avoid Unjustified 3D

- ▶ Depth judgment is poor
- ▶ Occlusion
- ▶ **Perspective Distortion**
- ▶ Colour: Lighting / Shadows / Shading
- ▶ Tilted Text illegible

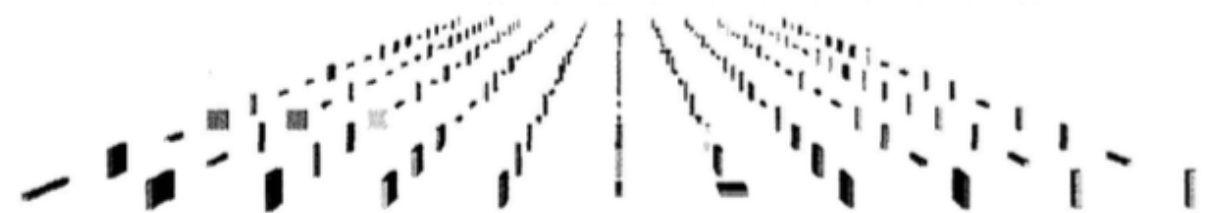
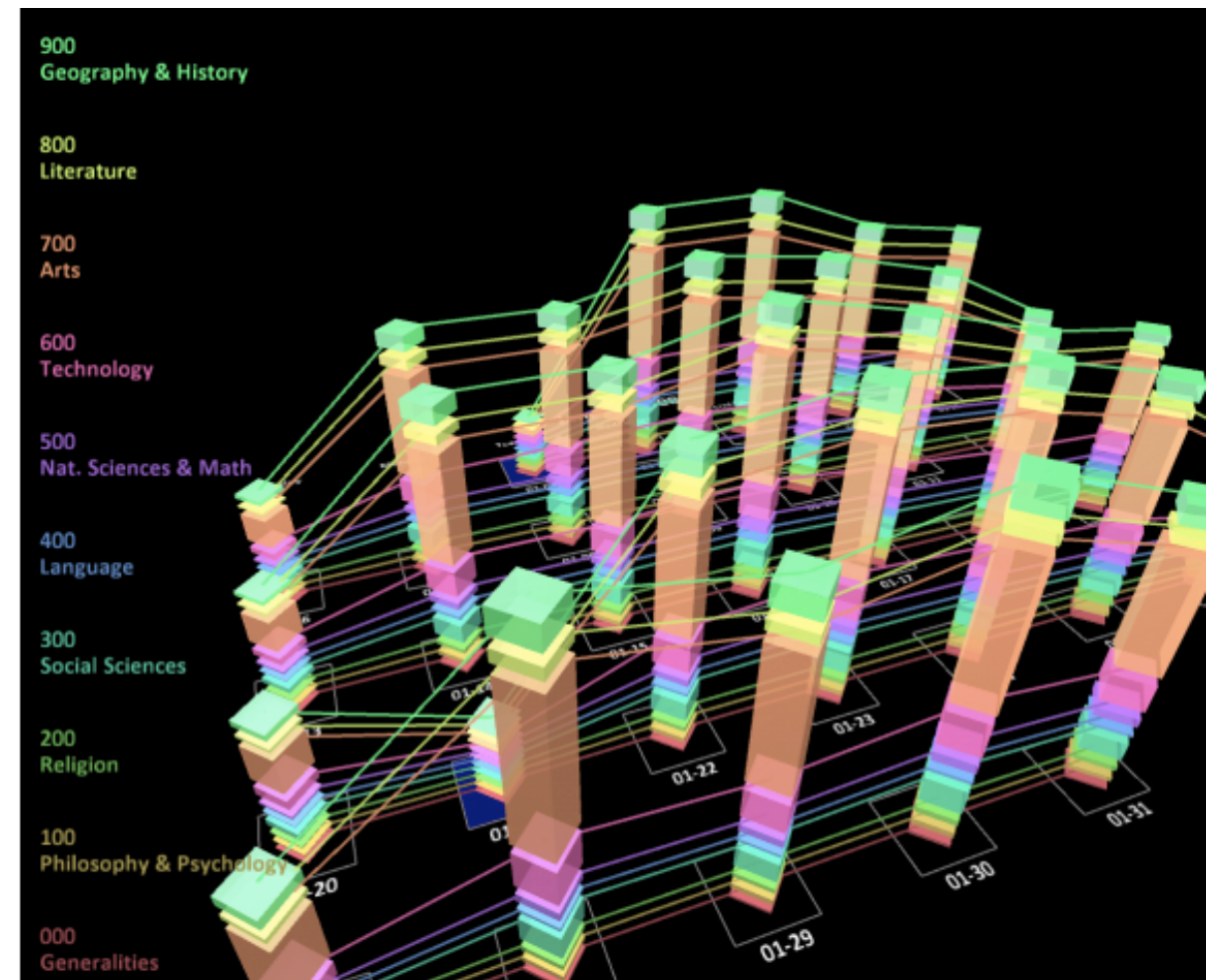
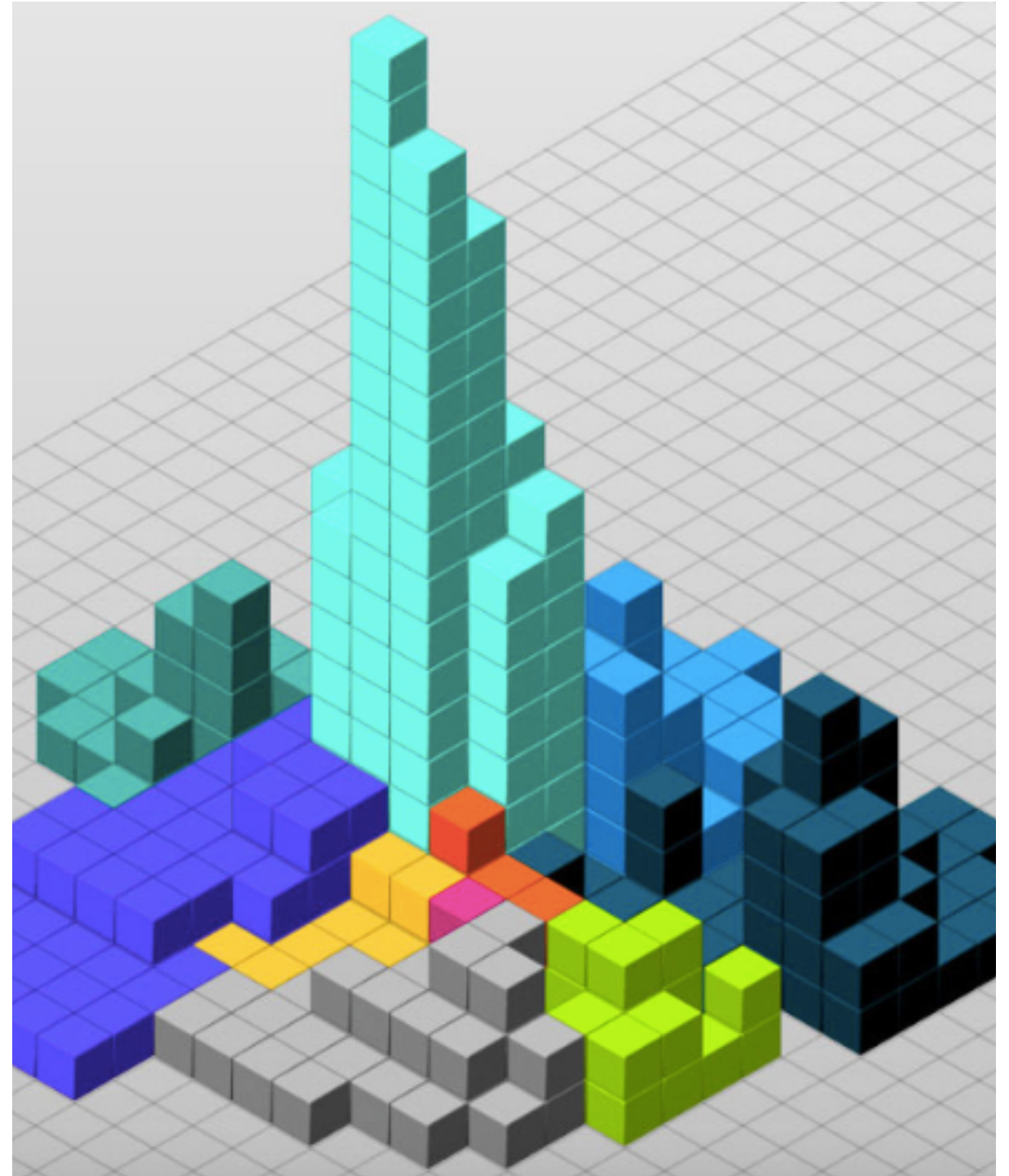


Figure 6.5. With perspective distortion, the power of the planar spatial position channel is lost, as is the size channel. From [Mukherjea et al. 96, Figure 1].

GUIDELINES – MUNZNER

Avoid Unjustified 3D

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DATA VISUALISATION

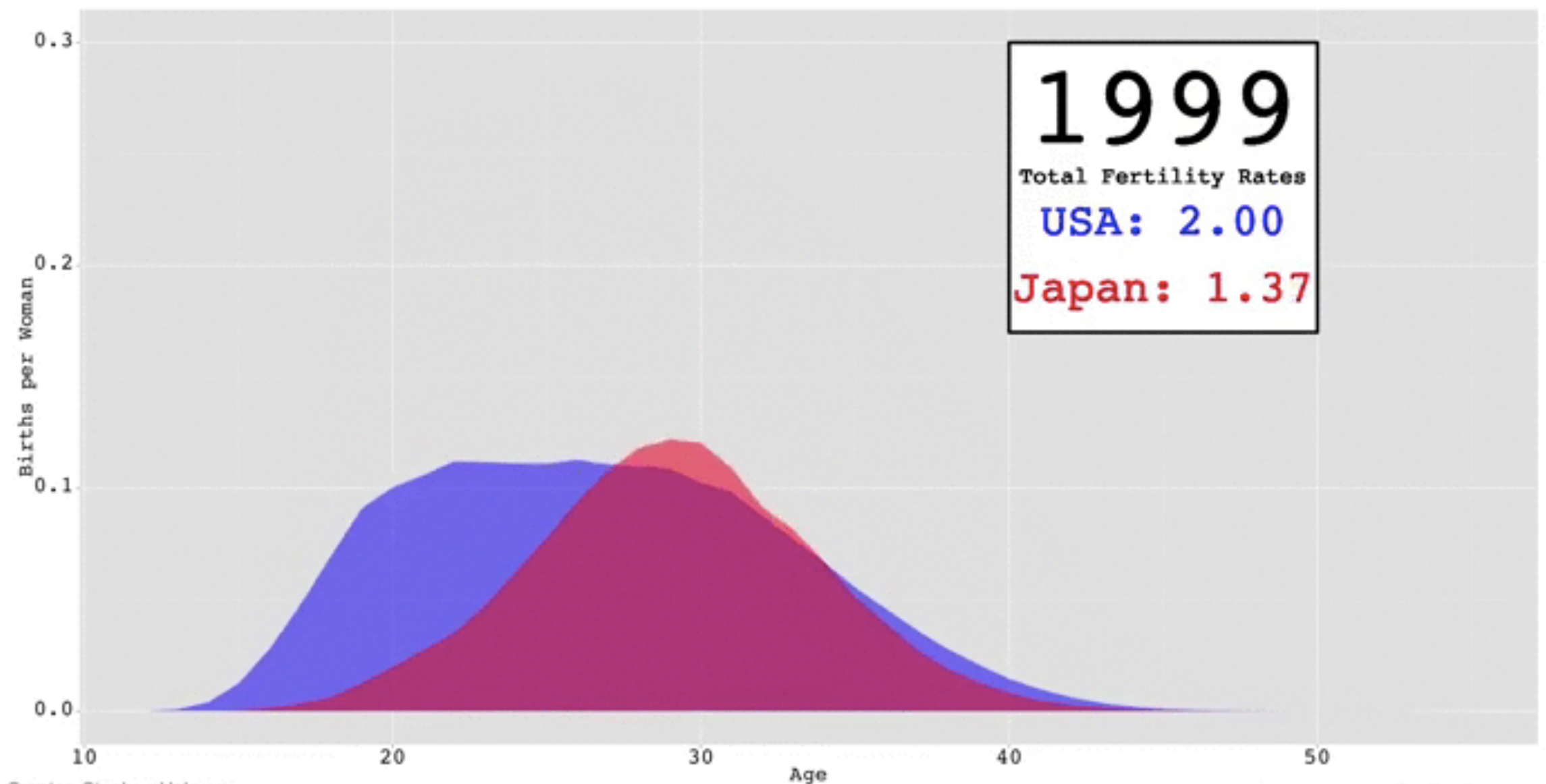
TOPIC 02-5: DESIGN GUIDELINES ANIMATIONS

GUIDELINES – MUNZNER

- ▶ Eyes beat memory
 - ▶ Cognitive limitations (working memory)
 - ▶ Animation vs Side by Side Views
 - ▶ Change blindness

ANIMATION

USA and Japan Fertility Over Time

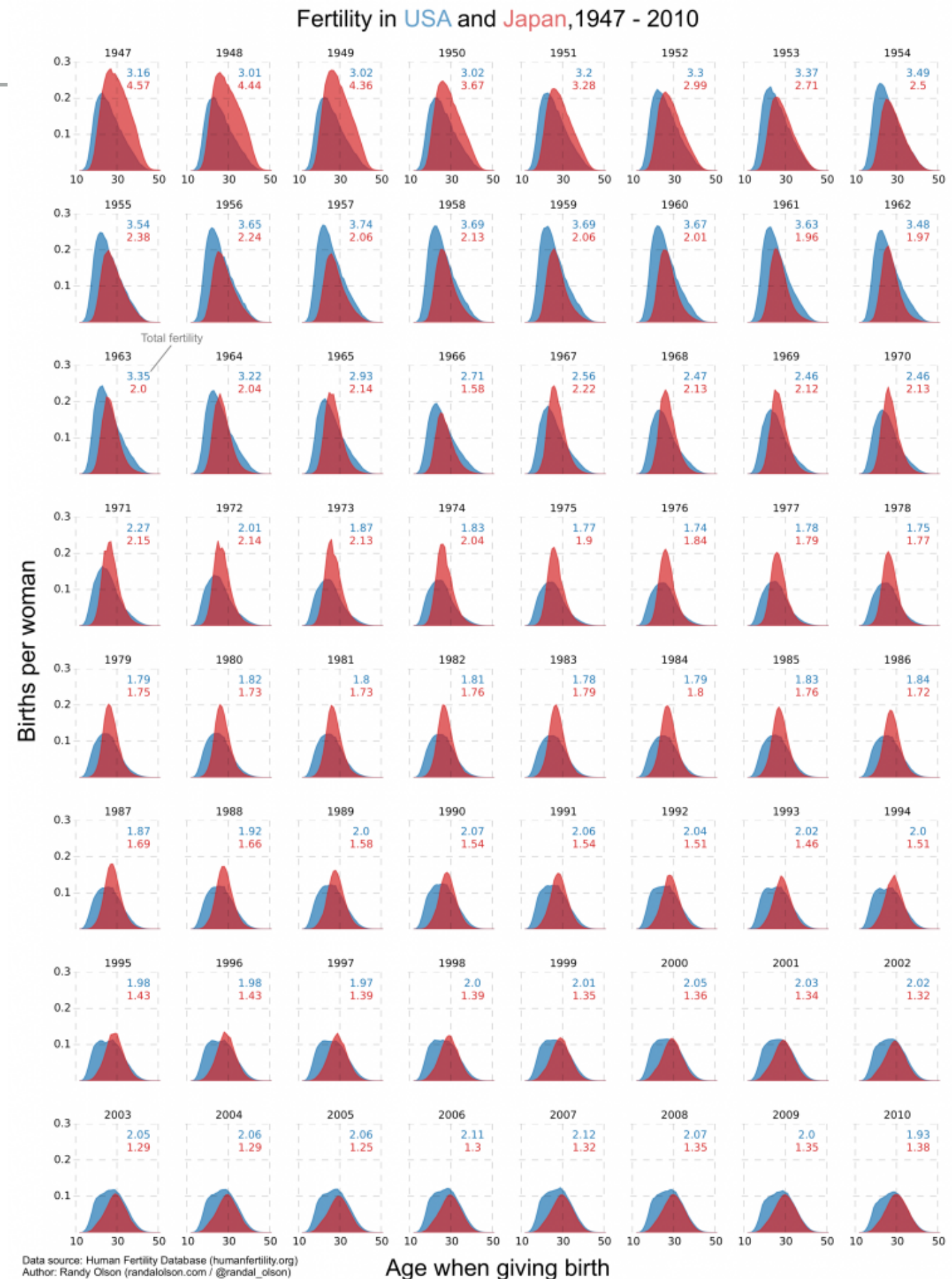


Creator: Stephen Holzman

Source: Human Fertility Database

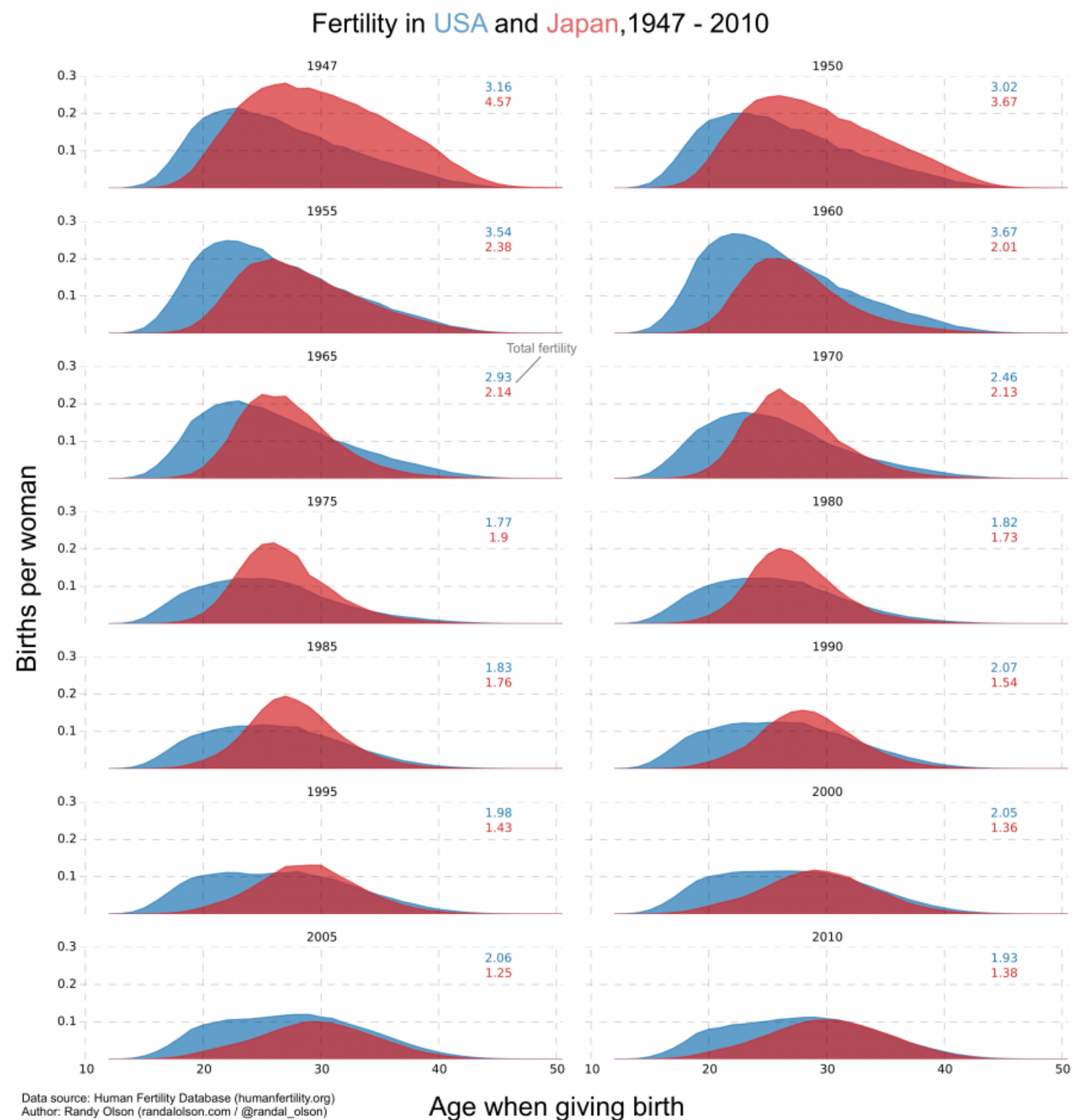
SMALL MULTIPLES

- ▶ data shown in small frames
- ▶ the same combination of variables...



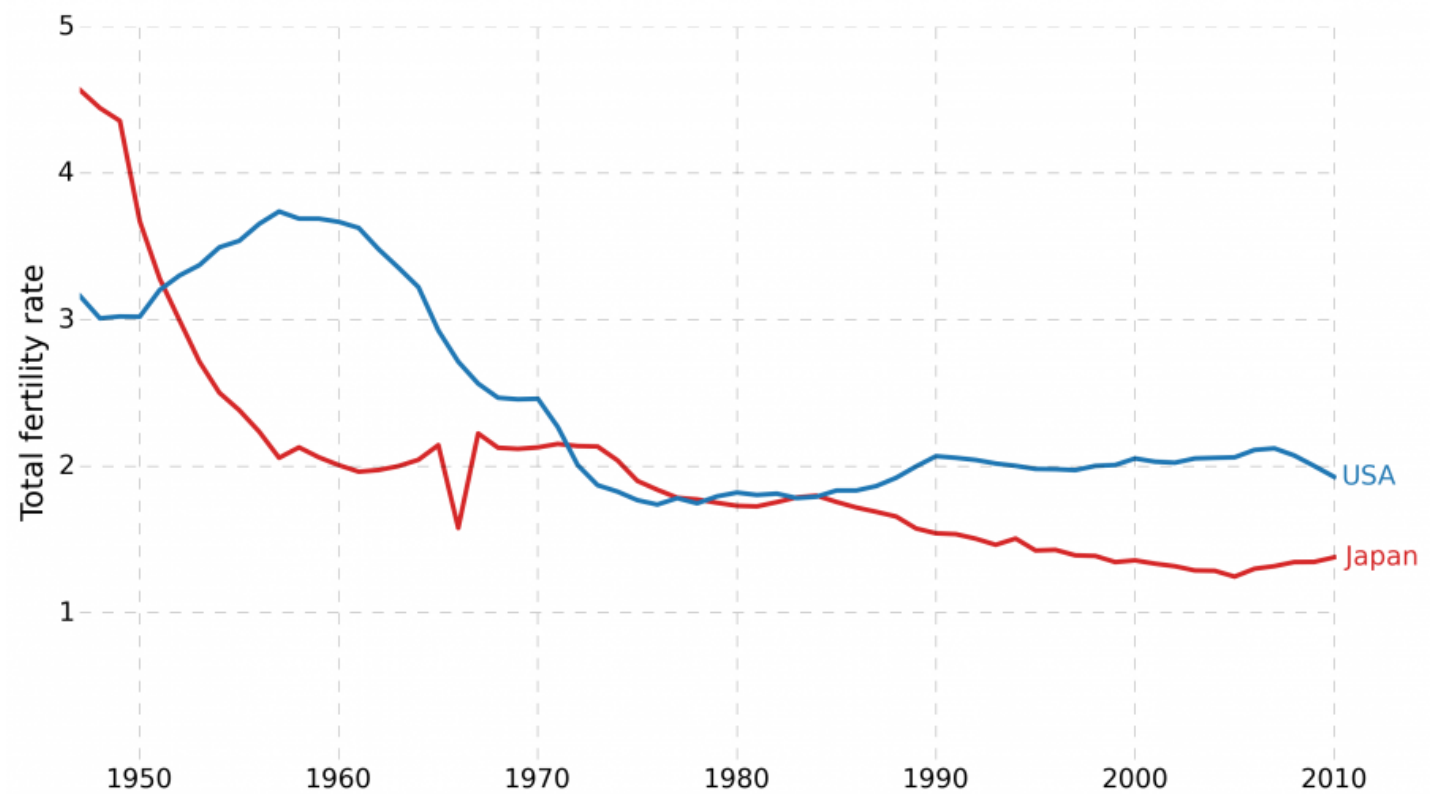
LESS SMALL MULTIPLES

- ▶ Make a judgement about how small the multiples need to be...



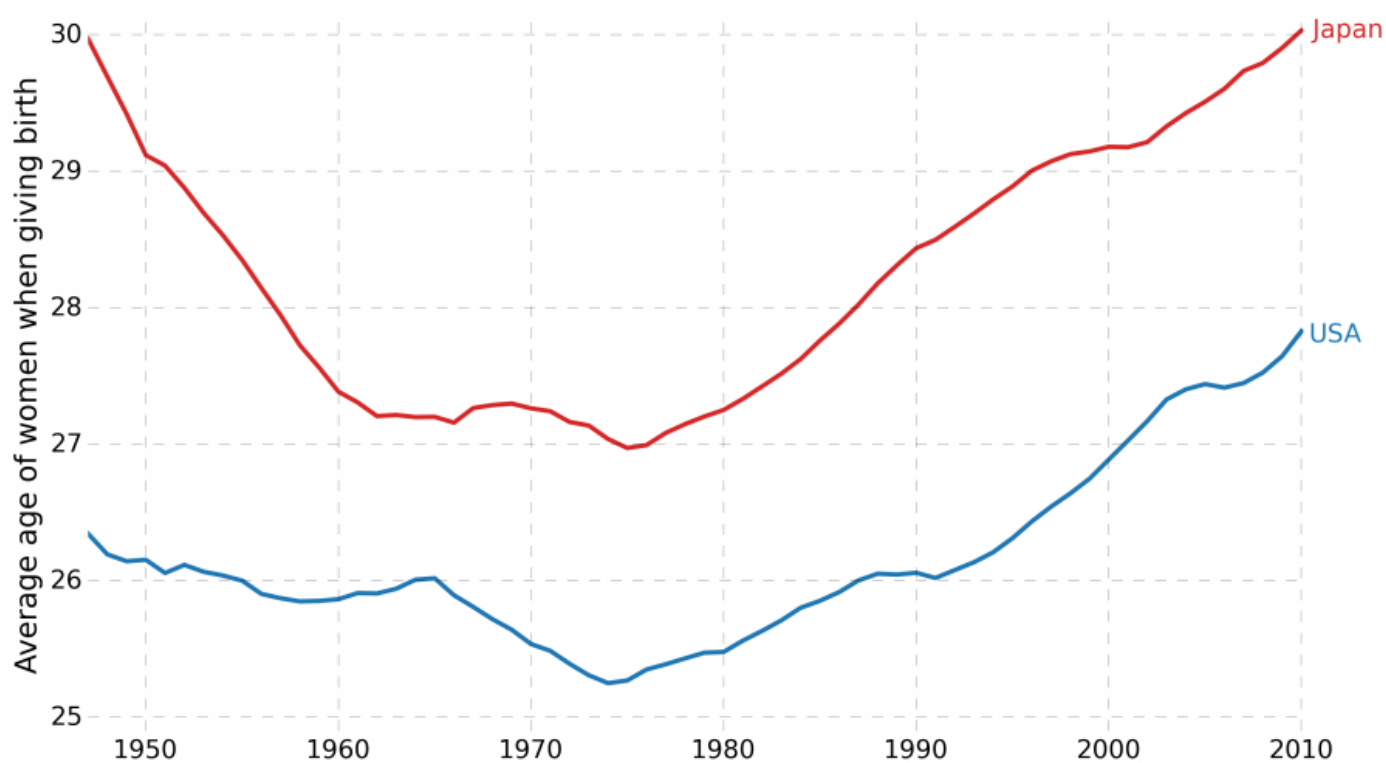
OLD FAITHFUL!

Total fertility rate in USA and Japan, 1947 - 2010



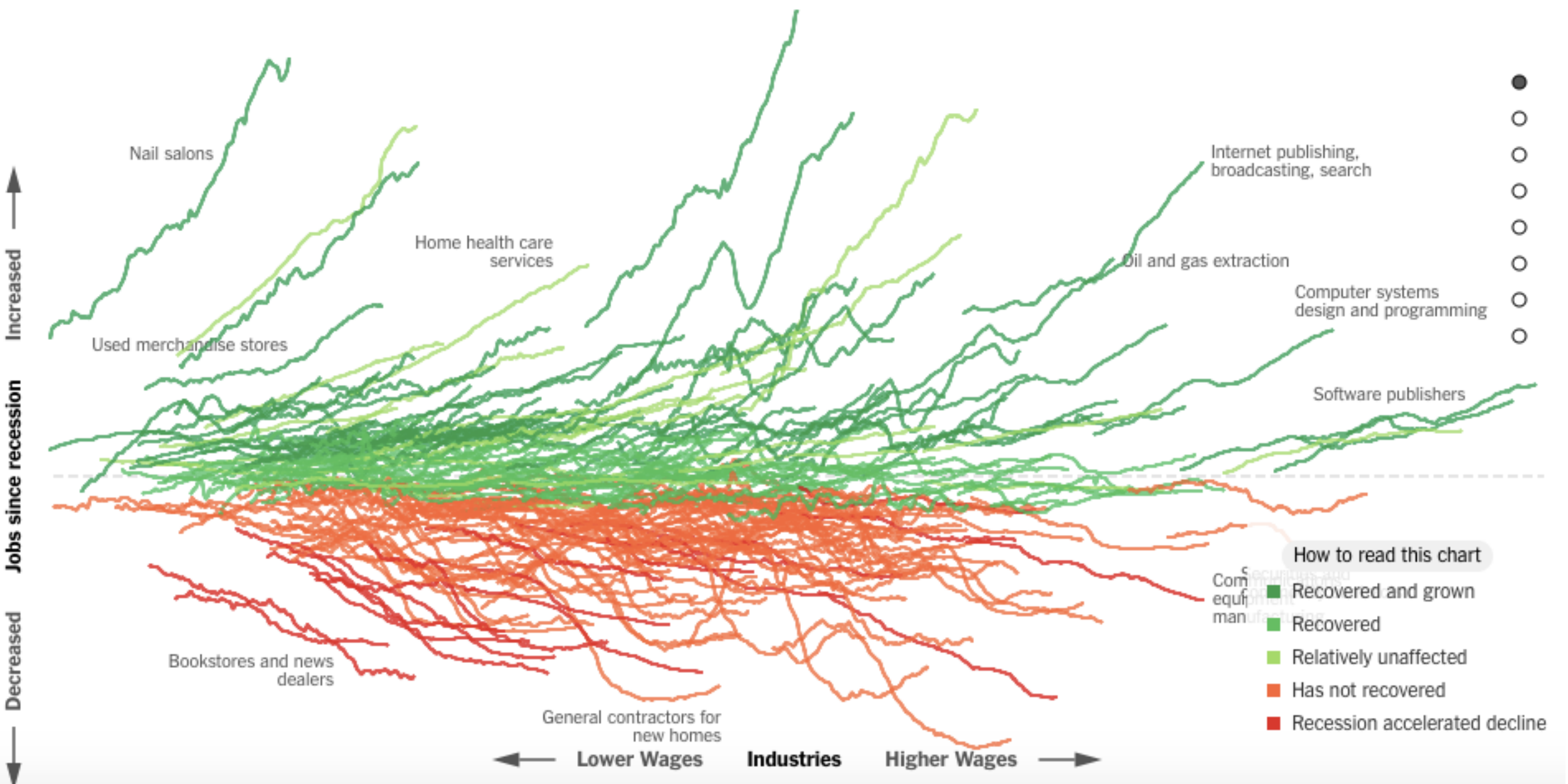
Da
Au

Average age when giving birth in USA and Japan, 1947 - 2010



SMALL MULTIPLES

How the Recession Reshaped the Economy, in 255 Charts





Photographic images
used in this presentation:

mostly from Microsoft Clipart circa 2009
(unless otherwise specified)
Note references are linked to images in PDF
version of this presentation.

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DATA VISUALISATION

TOPIC 02-6: BUBBLE CHART DISCUSSION

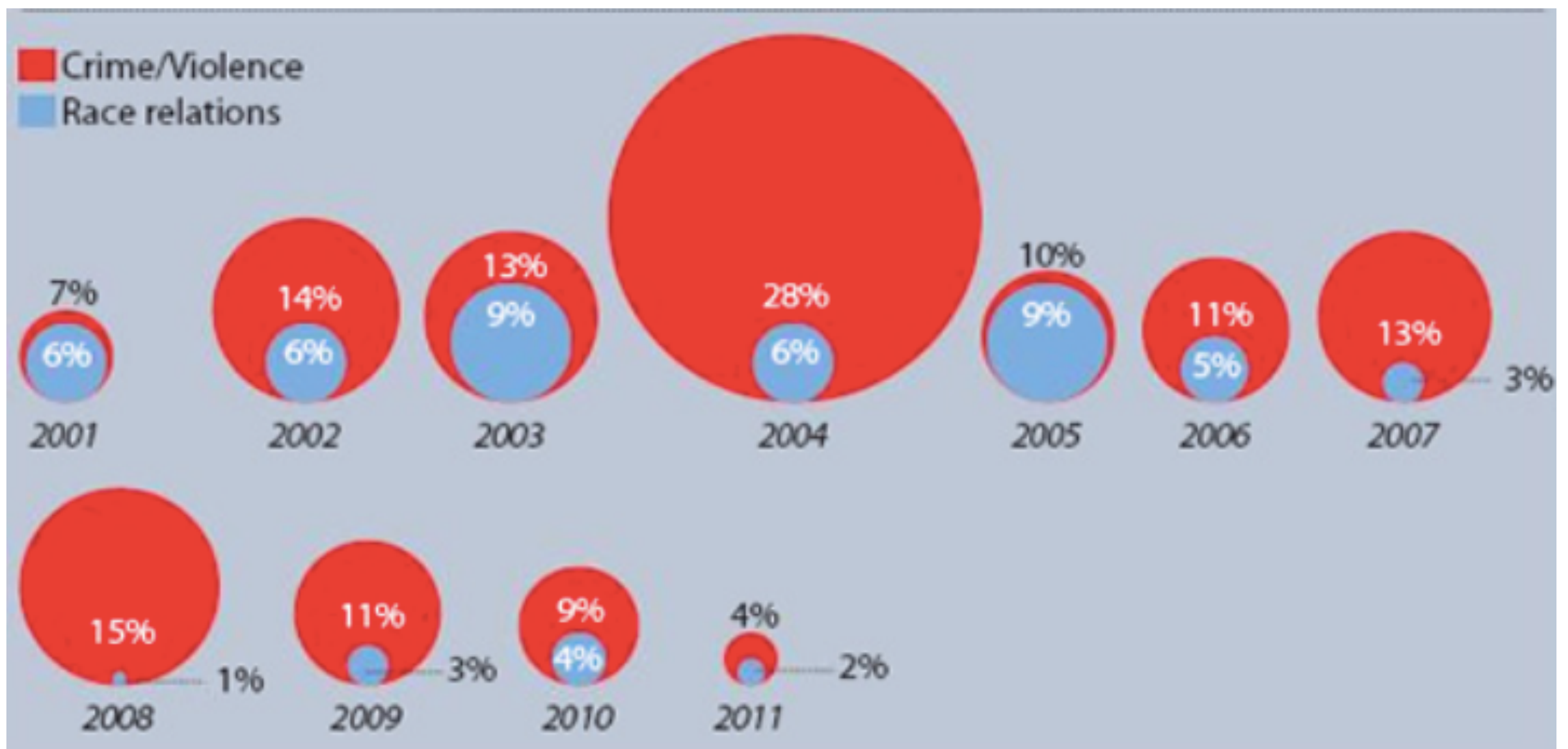
DESIGN CRITIQUE

- ▶ Who is the audience? (expert? non-expert?)
- ▶ What questions does this visualization answer?
- ▶ What design principles best describe why it is good / bad?
- ▶ Why do you like / dislike this visualization? Can you suggest any improvements?

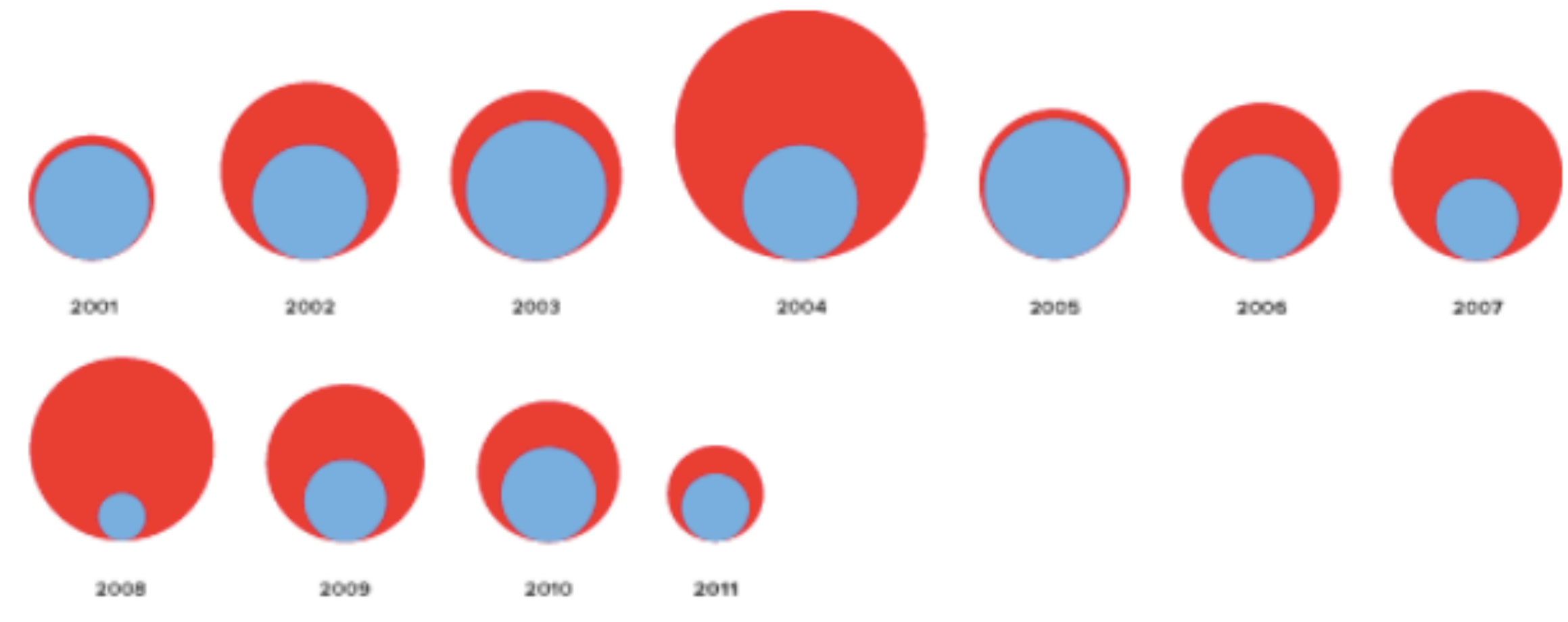
BUBBLE CHART – DIAMETER



BUBBLE CHART



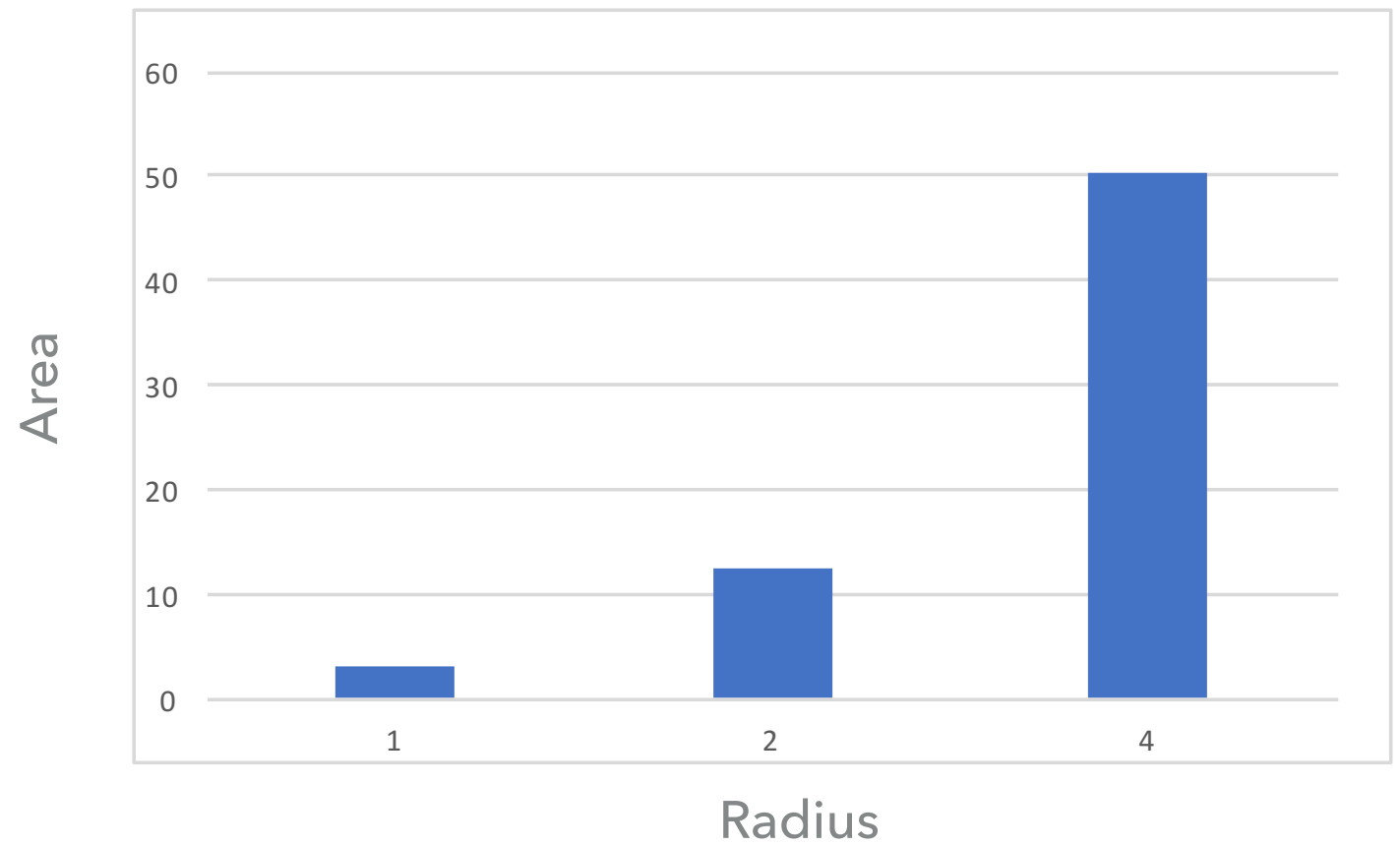
BUBBLE CHART – AREA CODING



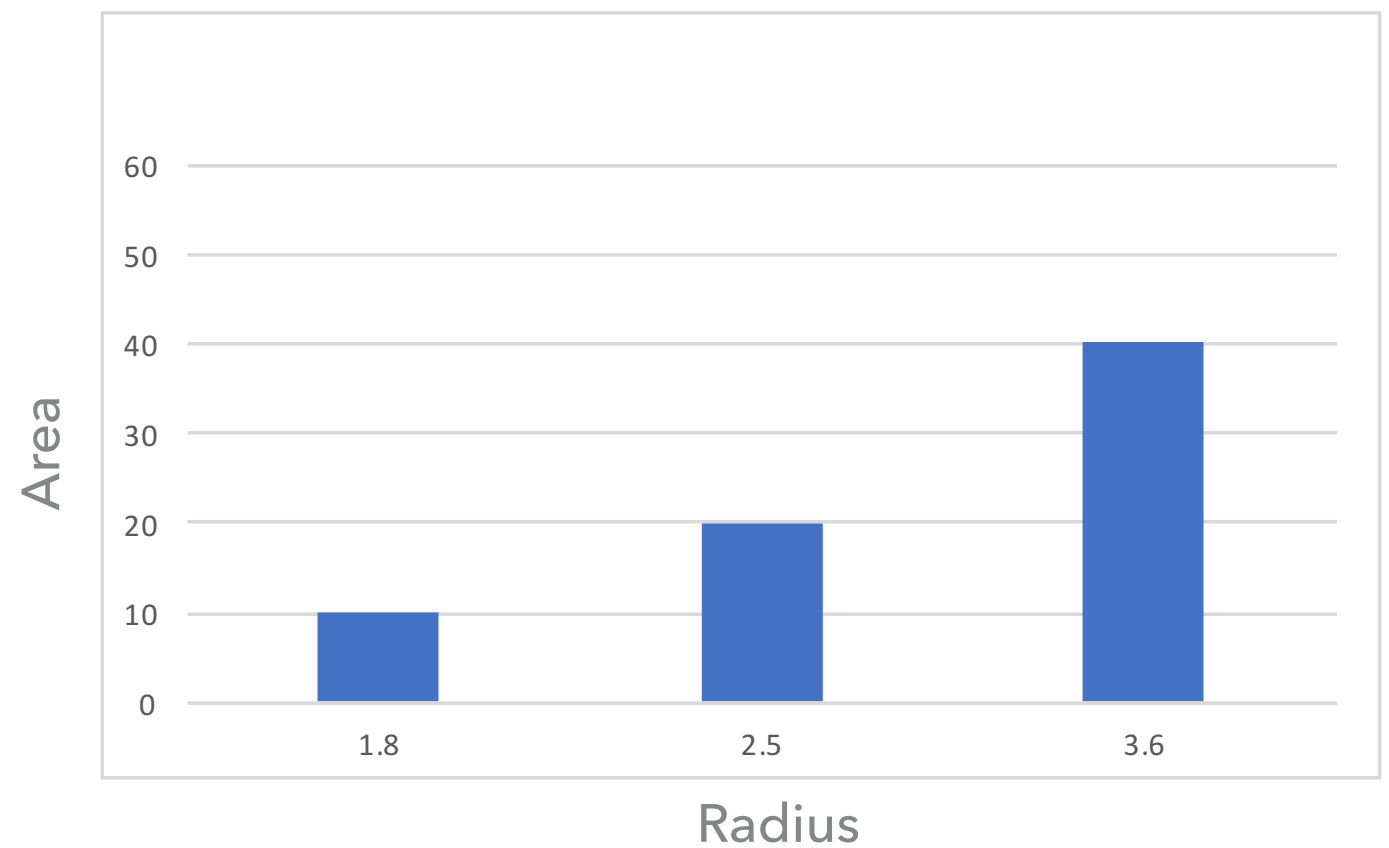
AREA vs RADIUS CODING



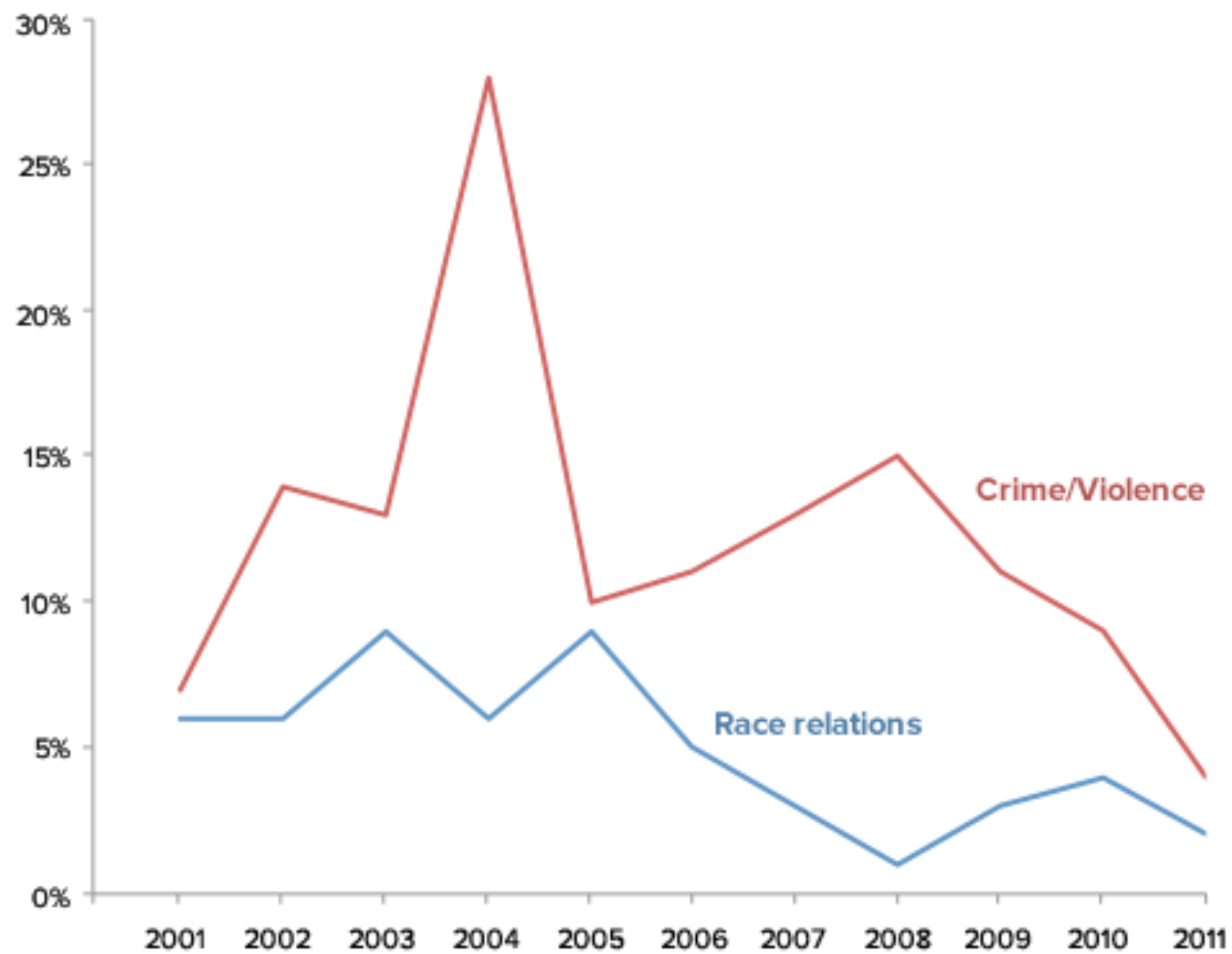
Double the radius



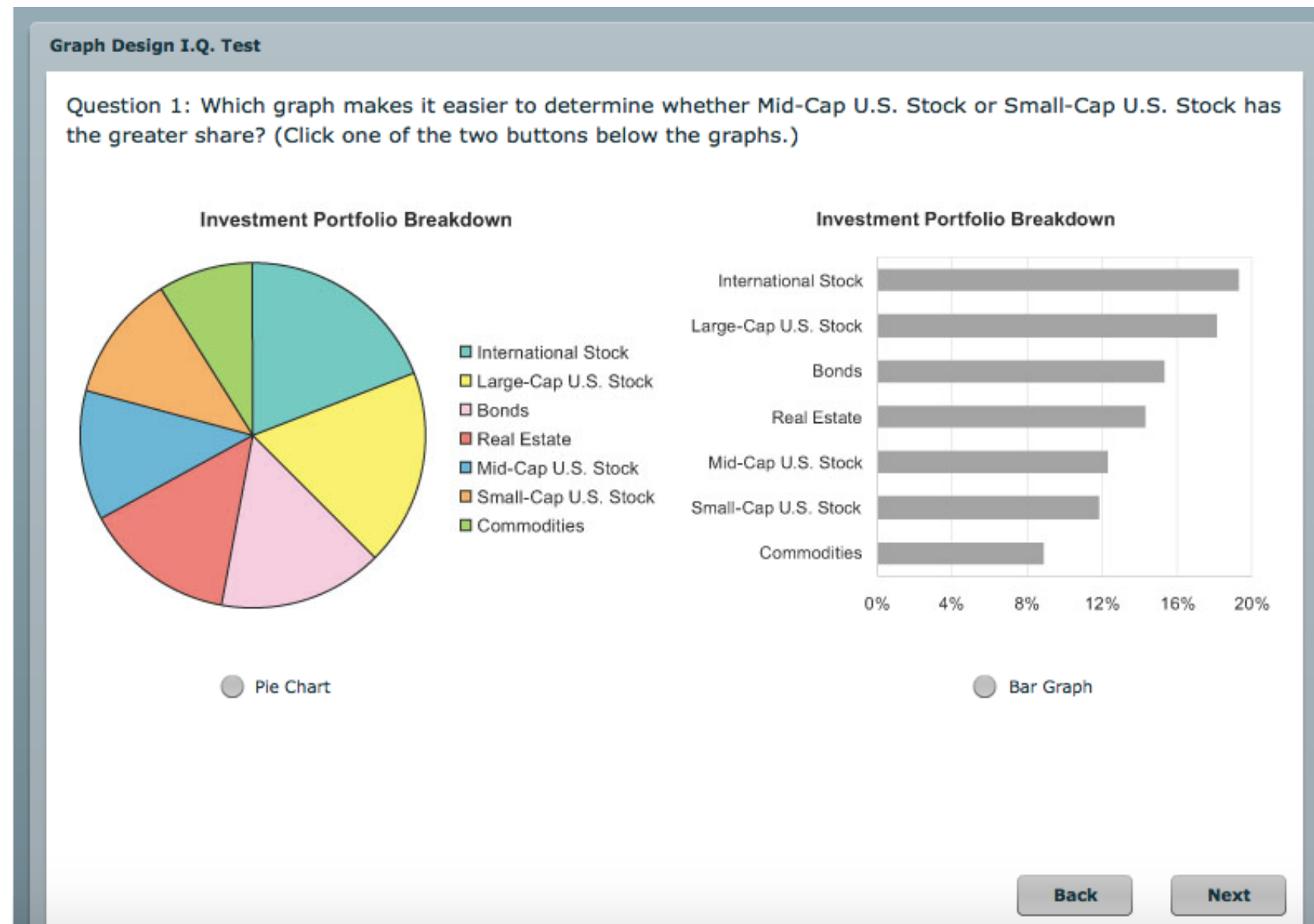
Double the area



LINE GRAPH



TEST YOUR CHART SENSE



<http://www.perceptualedge.com/files/GraphDesignIQ.html>

Clutter and confusion are not attributes of information, they are failures of design.

Edward Tufte

Statistician and data
visualisation guru





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